METRIC

MIL-STD-1893(AT)
14 August 1891
SUPERSEDING
(see 6.1)

MILITARY STANDARD

# CLEANING ELECTROCHEMICALLY MACHINED NICKEL BASE ALLOY SURFACES, PROCESS FOR (METRIC)

AMSC N/A AREA MFFP

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# FOREWORD

1. This Military Standard is approved for use by U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations; additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. The intent of this standard is to define the process used for removing contaminants from electrochemically machined surfaces of nickel base alloy parts.



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# 1. SCOPE

1.1 Scope. This standard defines the process used for removing contaminants from electrochemically machined surfaces of nickel base alloy parts.



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## 2. REFERENCED DOCUMENTS

(This section is not applicable to this standard.)



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## 3. DEFINITIONS

(This section is not applicable to this standard.)



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# 4. GENERAL REQUIREMENTS

4.1 Equipment.

4.1.1 Dipping racks. Plastic coated or stainless steel racks shall be used when immersing parts in decontamination solution.

CAUTION: Iron or low alloy steel racks and handling equipment shall not be used for immersion in solution.

4.2 Materials.

4.2.1 Degreaser solution. Stabilized 1-1-1 trichloroethane.

4.2.2 Decontamination solution. The solution shall be formulated using equal parts by volume of water and nitric acid ($HNO_3$, 40° Baume) plus 5 milliliters (0.17 ounce) of hydrochloric acid (HCl, concentrated) for each 3.78 liters (one gallon) of solution.



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## 5. DETAILED REQUIREMENTS

5.1 Procedure. The parts shall be cleaned as follows (see 4.1.1 for caution note):

a. Vapor degrease (see 4.2.1) parts paying special attention to cored passages and narrow openings.

b. Pressure flush passages and parts with clean cold running water of maximum 100 degrees Celsius (°C) [100 degrees Fahrenheit (°)F].

c. Immerse parts in decontamination solution (see 4.2.2) at room temperature for 50 ± 5 minutes.

d. Pressure flush passages and parts with clean cold running water [maximum 38°C (100°F)].

e. Rinse in hot water [54 - 60°C (120 - 140°F)].

f. Dry passages and parts with pressure blasts of filtered clean air.

g. If necessary, repeat steps a thru f to insure complete removal of surface contaminants.



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## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended to define the process for removing contaminants from electrochemically machined surfaces of nickel base alloy parts.

6.2 Supersession. This military standard supersedes AVCO Lycoming Division, internal specification P7207A, dated 27 March 1975.

6.3 Subject term (key word) listing.

Contaminant
Decontamination solution
Degreaser
Dripping racks
Electrochemical
Hydrochloric acid
Nickel base alloy
Nitric acid
Trichloroethane

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project MFFP-A491)



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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-STD-1893(AT); Cleaning Electrochemically Machined Base Alloy Surfaces, Process for (Metric)

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